

Statistical and Machine Learning

Exam — Exercise 1 (Mock version) — Solutions

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Marking. Each statement is worth **2 marks**: one for the correct True/False answer and one for a correct justification (total **20 marks**). Model justifications are shown in blue; award the justification mark for any response that captures the same essential reason.

Exercise 1 — True / False — Solutions (20 marks)

(a). **(Linear models)** Consider the ordinary least square regression with parameter $\beta \in \mathbb{R}^p$ estimated by $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ such that $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \mathbf{H}\mathbf{Y}$.

Answer: True. $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$

(b). **(Evaluation)** Consider the classification problem. The ROC curve plots the False Positive Rate on the x -axis against the True Positive Rate on the y -axis.

Statement: If the AUC (area under the curve) is 0.75, the classifier is worse than the random guessing.

Answer: False Plot or 0.5 = random guess, 1 = perfect classifier. Anything in between – better than random guess.

3. **(Regularisation & Bayesian)** For $\lambda > 0$, the ridge estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$ exists even when $\mathbf{X}^\top \mathbf{X}$ is singular.

Answer: True. Adding $\lambda \mathbf{I}$ with $\lambda > 0$ makes $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ positive definite and therefore invertible, even when $\mathbf{X}^\top \mathbf{X}$ is singular (e.g., when $p > n$).

4. **(Exemplar & kernel methods)** k -nearest-neighbour classification is a parametric method that assumes a specific functional form for the decision boundary.

Answer: False. k -NN is fully *nonparametric*: it makes no assumption about the shape of the decision boundary.

5. **(Trees & boosting)** In a regression tree, the prediction assigned to a terminal node (leaf) is the mean of the response values of the training observations falling in that node.

Answer: True. A regression tree predicts the constant $\hat{c}_m = \frac{1}{|R_m|} \sum_{i: x_i \in R_m} y_i$, i.e., the sample mean of the responses in the leaf.

6. (Clustering) Purity is a reliable measure of clustering quality because it penalises solutions that use a very large number of clusters.

Answer: False. Purity does *not* penalise using too many clusters — it is trivially equal to 1 when every observation forms its own cluster.

7. (Dimension reduction) In PCA, the first principal component is the direction that maximises the variance of the projected data.

Answer: True. Minimising the reconstruction error is equivalent to maximising the variance of the projection; the first principal component is the eigenvector of the empirical covariance matrix with the largest eigenvalue.

8. (Neural networks — tabular) A multilayer perceptron using only linear activation functions can represent a strictly larger class of functions than a single linear model.

Answer: False. A composition of linear layers is again a linear map ($\mathbf{W}'\mathbf{x}$), so a purely linear MLP represents exactly the same class of functions as a single linear model — nonlinear activations are essential.

9. (Neural networks — images) A convolutional layer typically has far fewer parameters than a fully connected layer applied to the same image, because its filter weights are small and shared across spatial locations.

Answer: True. A convolutional layer uses small filters whose weights are shared (tied) across all spatial locations, so it has far fewer parameters than a dense layer, and this weight sharing is what gives translation invariance.

10. (Neural networks — sequences) In a standard RNN, the hidden state $\mathbf{h}_t$ depends only on the current input $\mathbf{x}_t$ and not on the previous hidden state $\mathbf{h}_{t-1}$.

Answer: False. The hidden state is updated recurrently, $\mathbf{h}_t = \varphi(\mathbf{W}_{xh}[\mathbf{x}; \mathbf{y}_{t-1}] + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h)$, so it depends on $\mathbf{h}_{t-1}$; this dependence is what gives the RNN its memory.

Question 2. Linear models (5 marks)

(a) (3 marks) Consider the ordinary least square regression with parameter $\beta \in \mathbb{R}^p$ estimated by $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. Under the Gauss–Markov assumptions $\mathbb{E}(\varepsilon) = \mathbf{0}$ and $\text{Cov}(\varepsilon) = \sigma^2 \mathbf{I}_n$, derive $\mathbb{E}(\hat{\beta})$ and $\text{Cov}(\hat{\beta})$.

Answer. Write $\hat{\beta} = \mathbf{A}\mathbf{Y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\mathbf{Y} = \mathbf{X}\beta_0 + \varepsilon$. Then $\mathbb{E}(\hat{\beta}) = \mathbf{A}\mathbf{X}\beta_0 = \beta_0$ (unbiased), and $\text{Cov}(\hat{\beta}) = \mathbf{A}(\sigma^2 \mathbf{I})\mathbf{A}^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$.

(b) (2 marks) Give two reasons why a standard linear regression model is inappropriate for modelling if the outcome variable is binary, i.e. $Y \in \{0, 1\}$.

Answer. 1. The linear predictor $x^\top \beta$ can take any value in $\mathbb{R}$, including values outside $[0, 1]$, which are impossible for a probability.

2. The errors $\varepsilon_i = Y_i - \pi_i$ cannot be homoscedastic: for a Bernoulli variable, $\text{Var}(Y_i | x_i) = \pi_i(1 - \pi_i)$, which depends on x_i .

Question 3. Evaluation (5 marks)

(a) (3 marks) A binary classifier gives $TP = 40$, $FP = 10$, $FN = 20$, $TN = 30$. Compute the accuracy and the precision and give one example of situation where precision would be an important metric to report.

Answer. Accuracy = $(TP + TN)/n = (40 + 30)/100 = 0.70$. Precision = $TP/(TP + FP) = 40/50 = 0.80$. Spam detection – of all emails flag are spam, how many are actual spam.

(b) (2 marks) Consider a K-fold cross-validation. Give one method to select the number of folds and explain what is the purpose of holding one fold at a time?

Answer. $K = 5$ or $K = 10$ are commonly used, as they balance bias against computational cost). The purpose of holding out data is to simulate how the model will perform on unseen observations. Because the held-out fold was not used for training, the prediction error it yields is an approximately unbiased estimate of the true generalisation error — unlike the training error, which systematically underestimates it due to overfitting

Question 4. Regularisation & Bayesian (5 marks)

(a) (2 marks) Write the ridge-regression penalised least-squares objective $PLS(\beta)$ and give the closed-form ridge estimator.

Answer. $PLS(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta) + \lambda \beta^\top \beta$; its minimiser is $\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top \mathbf{Y}$.

(b) (3 marks) For orthonormal covariates ($\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$), show that ridge scales each OLS coefficient by $\frac{1}{1+\lambda}$, and describe the limits $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.

Answer. With $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$, $\hat{\beta}_{\text{ridge}} = (\mathbf{I} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y} = \frac{1}{1+\lambda} \mathbf{X}^\top \mathbf{Y} = \frac{1}{1+\lambda} \hat{\beta}_{\text{LS}}$, so $|\hat{\beta}_{j,\text{ridge}}| = \frac{1}{1+\lambda} |\hat{\beta}_{j,\text{LS}}|$. As $\lambda \rightarrow 0$, $\hat{\beta}_{\text{ridge}} \rightarrow \hat{\beta}_{\text{LS}}$; as $\lambda \rightarrow \infty$, $\hat{\beta}_{\text{ridge}} \rightarrow \mathbf{0}$.

Question 5. Exemplar & kernel methods (5 marks)

(a) (2 marks) Write the k -NN estimate of the probability that a point $\mathbf{x}$ belongs to class l , and define every symbol.

Answer. $\hat{P}(Y = l | \mathbf{x}) = \frac{1}{k} \sum_{i \in N_x^k} \mathbb{I}\{y_i = l\}$, where N_x^k is the set of the k training points nearest to $\mathbf{x}$ (under the chosen distance) and the sum counts how many of them carry label l .

(b) (3 marks) For $k = 5$, a query point has nearest neighbours with labels $\{A, A, B, A, C\}$. Give the predicted class and $\hat{P}(Y = A | \mathbf{x})$, then state one advantage and one disadvantage of k -NN relative to logistic regression.

Answer. Majority class = A (3 of 5), so $\hat{P}(Y = A | \mathbf{x}) = 3/5 = 0.6$. Advantage: fully nonparametric, so it can fit highly nonlinear boundaries. Disadvantage: it produces no coefficients, so it does not indicate which predictors are important.

Question 6. Trees & boosting (5 marks)

(a) (4 marks) A two-class node contains 10 observations: 6 of class A and 4 of class B ($\hat{\pi}_A = 0.6$, $\hat{\pi}_B = 0.4$). Compute the Gini index and the misclassification error of the node.

Answer. Gini $i_2 = 1 - \sum_k \hat{\pi}_k^2 = 1 - (0.6^2 + 0.4^2) = 1 - 0.52 = 0.48$. Misclassification $i_1 = 1 - \max_k \hat{\pi}_k = 1 - 0.6 = 0.4$.

(b) (1 marks) What do the above metrics measure?

Answer. Both metrics measure node impurity, with 0 indicating a pure node (single class) and higher values indicating more evenly mixed classes. They are used to guide tree splitting. Gini is generally preferred over misclassification error for growing trees since it is more sensitive to changes in class proportions.

Question 7. Clustering (5 marks)

(a) (3 marks) Describe the two steps of the K -means algorithm (assignment and update) with formulas, and state the objective (distortion) it minimises.

Answer. Assignment: $z_n^* = \arg \min_k \|\mathbf{x}_n - \boldsymbol{\mu}_k\|_2^2$. Update: $\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n: z_n=k} \mathbf{x}_n$. These alternately minimise the distortion $J = \sum_n \|\mathbf{x}_n - \boldsymbol{\mu}_{z_n}\|^2$ (to a local minimum).

(b) (2 marks) Explain why the number of clusters K cannot be chosen by minimising the distortion J on the data, and name one principled alternative.

Answer. J decreases monotonically with K and is trivially 0 when $K = N$, so minimising it always selects the largest K . Alternatives: the silhouette score (maximise it), the elbow heuristic, or the BIC/marginal likelihood of a Gaussian mixture.

Question 8. Dimension reduction (5 marks)

(a) (3 marks) Explain the encode/decode view of PCA and write the reconstruction error (distortion) being minimised; state what the optimal $\mathbf{W}$ is in terms of the covariance matrix.

Answer. Encode $\mathbf{z} = \mathbf{W}^\top \mathbf{x}$, decode $\hat{\mathbf{x}} = \mathbf{W}\mathbf{z}$; minimise $\mathcal{L}(\mathbf{W}) = \frac{1}{N} \sum_n \|\mathbf{x}_n - \mathbf{W}\mathbf{W}^\top \mathbf{x}_n\|_2^2$ over orthonormal $\mathbf{W}$. The optimum is $\hat{\mathbf{W}} = \mathbf{U}_L$, the L eigenvectors of the empirical covariance $\hat{\boldsymbol{\Sigma}}$ with the largest eigenvalues.

(b) (2 marks) State the equivalence between minimising reconstruction error and maximising variance, and hence what the first principal component maximises.

Answer. Minimising the reconstruction error is equivalent to maximising the variance of the projection. The first principal component is the unit vector $\mathbf{w}_1$ maximising $\mathbf{w}_1^\top \hat{\boldsymbol{\Sigma}} \mathbf{w}_1$ (the top eigenvector); the maximised value equals its eigenvalue λ_1 .

Question 9. Neural networks — tabular (5 marks)

(a) (2 marks) Explain why an MLP needs nonlinear activation functions: what happens if every activation is linear?

Answer. A composition of linear layers is itself linear: $\mathbf{W}_L \cdots \mathbf{W}_1 \mathbf{x} = \mathbf{W}' \mathbf{x}$. So a purely linear MLP is no more expressive than a single linear layer; nonlinear activations are what allow it to represent nonlinear functions.

(b) (3 marks) Give the sigmoid $\sigma(a)$ and its derivative, explain the vanishing-gradient problem it causes, and name one activation that mitigates it and why.

Answer. $\sigma(a) = \frac{1}{1+e^{-a}}$, $\sigma'(a) = \sigma(a)(1 - \sigma(a))$. For large $|a|$ the sigmoid saturates ($\sigma \rightarrow 0$ or 1) so $\sigma' \approx 0$; multiplied over many layers the gradient vanishes and early layers barely learn. $\text{ReLU}(a) = \max(a, 0)$ has derivative 1 for $a > 0$, so it does not saturate for positive inputs and preserves the gradient.

$$\sigma'(a) = -(1 + e^{-a})^{-2} \cdot \frac{d}{da} (1 + e^{-a}) = -(1 + e^{-a})^{-2} \cdot (-e^{-a}) = \frac{e^{-a}}{(1 + e^{-a})^2}$$

$$\frac{1 \times e^{-a}}{(1 + e^{-a})(1 + e^{-a})} = \sigma(a)(1 - \sigma(a))$$

Question 10. Neural networks — images (5 marks)

(a) (3 marks) Give the output-size formula (one spatial dimension) for a convolution with input size x , filter size f , zero-padding p per side and stride s ; then compute the output size for $x = 7$, $f = 3$, $p = 1$, $s = 2$.

Answer. Output size = $\lfloor \frac{x+2p-f}{s} \rfloor + 1$. For $x = 7, f = 3, p = 1, s = 2$: $\lfloor (7 + 2 - 3)/2 \rfloor + 1 = \lfloor 6/2 \rfloor + 1 = 3 + 1 = 4$.

(b) (2 marks) In what sense is a convolutional layer a special case of a fully connected (MLP) layer? Give the two structural properties of its weight matrix and what they buy you.

Answer. Its weight matrix is sparse (most entries zero) and tied (the nonzero weights are shared across all spatial locations) — a Toeplitz-like structure. This gives translation invariance and drastically fewer parameters than a dense layer.

Question 11. Neural networks — sequences (5 marks)

(a) (3 marks) Write the RNN hidden-state update $\mathbf{h}_t$ (in terms of the input, previous output, and previous hidden state) and explain why $\mathbf{y}_t$ implicitly depends on all past observations.

Answer. $\mathbf{h}_t = \varphi(\mathbf{W}_{xh}[\mathbf{x}; \mathbf{y}_{t-1}] + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h)$. Since $\mathbf{y}_t$ depends on $\mathbf{h}_t$, which depends on $\mathbf{h}_{t-1}$ (hence on $\mathbf{y}_{t-2}$), and so on recursively, $\mathbf{y}_t$ depends on all earlier inputs/outputs; the hidden state acts as memory.

(b) (2 marks) What is backpropagation through time (BPTT), and why is truncated BPTT commonly used?

Answer. BPTT applies backpropagation to the RNN computation graph unrolled over time to obtain parameter gradients. The full gradient sums over all past steps (cost $O(T^2)$) and can vanish or explode; truncated BPTT keeps only the most recent K steps, cutting cost and improving stability.

Question 12. Synthesis: Linear models, Regularisation & Bayesian, Evaluation (10 marks)

(a) (6 marks) Consider ridge regression with tuning parameter λ . (i) Using the bias-variance decomposition, explain how increasing λ affects the bias and the variance of $\hat{\beta}_{\text{ridge}}$, and describe the two limiting cases $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$. (ii) Describe how K -fold cross-validation is used to select λ

in practice, and explain why choosing λ by minimising the *training* error alone would always lead to selecting $\lambda = 0$.

Answer. (i) As λ increases, $\hat{\beta}_{\text{ridge}}$ is shrunk further toward $\mathbf{0}$: this increases bias (the estimator moves away from the unbiased OLS solution) but decreases variance (shrinkage dampens the estimator's sensitivity to the noise in $\mathbf{Y}$). At $\lambda = 0$, ridge coincides with OLS: unbiased but highest variance; as $\lambda \rightarrow \infty$, $\hat{\beta}_{\text{ridge}} \rightarrow \mathbf{0}$: maximal bias but zero variance. (ii) For a grid of candidate λ values, fit the ridge model on $K - 1$ folds and evaluate its error on the held-out fold, repeat for each fold, and average to get $CV(\lambda)$; choose the λ minimising $CV(\lambda)$. Training error decreases monotonically as $\lambda \rightarrow 0$ (less penalty always fits the training data at least as well), so minimising it alone always favours $\lambda = 0$ (OLS), ignoring the variance cost and leading to overfitting; only an estimate of *test* error, such as CV, can properly balance the bias–variance trade-off.

(b) (4 marks) Give a real-world regression scenario with many correlated predictors (e.g. gene-expression or economic data) and explain, using the bias–variance trade-off, why you would prefer a regularised approach over plain OLS. If your goal is also automatic variable selection, which regulariser would you choose, and why?

Answer. Example: predicting a health outcome from thousands of correlated gene-expression measurements with $p \gg n$. Here $\mathbf{X}^\top \mathbf{X}$ is ill-conditioned or singular, so OLS is unbiased but has huge variance (or does not exist), giving poor test performance. Ridge (or lasso) trades a small amount of bias for a large reduction in variance, improving generalisation. If automatic variable selection is also wanted, lasso is preferable, since its ℓ_1 penalty can set coefficients exactly to zero, yielding a sparse, more interpretable model (or elastic net if predictors form correlated groups, since lasso alone tends to arbitrarily pick one variable per group).

Question 13. Synthesis: Trees & boosting, Neural networks (tabular) (10 marks)

(a) (6 marks) (i) Random forests reduce variance by averaging many de-correlated trees; deep neural networks are often regularised using dropout. Explain the conceptual similarity between these two techniques as implicit forms of model averaging. (ii) Give one structural reason a fully connected deep neural network may overfit more easily than a tree ensemble on a tabular dataset with only a few thousand observations.

Answer. (i) Both can be seen as training an implicit ensemble and averaging over it: a random forest explicitly averages predictions from trees fit to bootstrap samples with random feature subsets at each split; dropout randomly deactivates units on each training pass, so training approximates fitting an ensemble of exponentially many weight-sharing sub-networks, which is then approximated at test time by using the full network (with rescaled activations). Both trade a single high-variance predictor for an averaged/ensembled one, reducing variance. (ii) A dense network has a very large number of freely-adjustable weights (every unit connects to every unit in the next layer), so its capacity relative to a few thousand observations is high and, without strong regularisation, it can memorise the training data; a tree ensemble's individual trees make simple axis-aligned splits and averaging many bootstrapped, de-correlated trees keeps the ensemble's variance controlled even with modest data.

(b) (4 marks) You are asked to build a fraud-detection model on a tabular dataset with mixed numeric and categorical features and only around 5,000 labelled examples. Would you recommend a gradient-boosted tree ensemble or a deep MLP as a first approach? Justify your choice with reference to sample size, interpretability, and regularisation.

Answer. Gradient-boosted trees (e.g. XGBoost) would typically be recommended. With only $\sim 5,000$ examples, tree ensembles usually generalise better than deep MLPs on tabular/mixed-type data, require less feature preprocessing (they handle categorical splits and are insensitive to feature scaling), and have well-understood regularisation controls (tree depth, learning rate/shrinkage, sub-sampling, number of rounds tuned via CV) that work well at this sample size. They are also more interpretable via feature importances, which matters for auditability in fraud detection. A deep MLP would likely need substantially more data and more careful tuning of regularisation (dropout, weight decay, early stopping) to match this performance.

Question 14. Synthesis: Dimension reduction, Clustering (10 marks)

(a) (6 marks) (i) Explain why applying PCA before running K -means on high-dimensional data is often beneficial, referencing both the reconstruction-error view of PCA and the distortion objective of K -means. (ii) The Gaussian mixture model (GMM) fitted by EM can be viewed as a probabilistic generalisation of K -means. State one assumption on the mixture components under which the GMM's E-step responsibilities reduce to the hard 0/1 assignments used by K -means.

Answer. (i) High-dimensional data suffers from the curse of dimensionality: distances become less discriminative and noisy or redundant dimensions can dominate the Euclidean distance used in K -means. PCA finds the L directions of maximal variance (equivalently, minimal reconstruction error $\frac{1}{N} \sum_n \|\mathbf{x}_n - \mathbf{W}\mathbf{W}^\top \mathbf{x}_n\|_2^2$), projecting the data onto a lower-dimensional subspace that retains most of the signal while discarding noise; distances in this reduced space are more meaningful for K -means' distortion objective $J = \sum_n \|\mathbf{x}_n - \boldsymbol{\mu}_{z_n}\|^2$, and computation is also cheaper. (ii) If every component shares the same isotropic covariance $\sigma^2 \mathbf{I}$ and $\sigma^2 \rightarrow 0$, the responsibilities r_{nk} concentrate entirely on the nearest cluster mean, becoming hard 0/1 assignments identical to K -means.

(b) (4 marks) Describe a short (2–3 step) analysis pipeline you would use to explore structure in a high-dimensional gene-expression dataset (thousands of genes, hundreds of samples) with no known labels, combining a dimension-reduction technique and a clustering technique. State briefly what each step contributes.

Answer. (1) Apply PCA to reduce thousands of genes to, say, the top 10–50 principal components, denoising the data and removing redundant/correlated dimensions while retaining most of the variance. (2) Run K -means, hierarchical clustering, or a GMM on the PCA-reduced representation to group samples into candidate clusters (e.g. putative cell types or disease subtypes), now working with more meaningful distances. (3) Optionally use t-SNE on the PCA-reduced data purely to visualise the discovered clusters in 2D (not to perform the clustering itself, since t-SNE does not preserve global distances reliably). Contribution: PCA denoises/compresses; the clustering step discovers group structure; t-SNE aids visual communication of the result.